

MOSAIC: Certifiable Stochastic Representation Release under Structured Deployment Shift Supplementary Material

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Code and artifacts: <https://github.com/RudraChopra/vera-edit-or-abstain/tree/mosaic-aaai2027-v1>

1 Notation and Exact Support Functions

For one task label, let $P = (p_1, \dots, p_G)$ collect the conditional fine-token laws and let $M \in \text{Stoch}(K, L)$ be a source-blind release channel. For an attacker assignment $a \in [G]^L$, define

$$v_{M,a,s}(c) = \sum_{z=1}^L M(c, z) \mathbf{1}\{a(z) = s\}. \quad (1)$$

Every $v_{M,a,s}$ lies in $[0, 1]^K$. The balanced Bayes accuracy has the two equivalent forms

$$\begin{aligned} A(P, M) &= \frac{1}{G} \sum_z \max_s (p_s M)(z) \\ &= \frac{1}{G} \max_{a \in [G]^L} \sum_s p_s^\top v_{M,a,s}. \end{aligned} \quad (2)$$

The second identity follows because a finite-alphabet Bayes rule can be chosen deterministically at each released token.

For empirical law $\hat{p}$, radius ϵ , and bounded score v , the confidence-set support function is

$$\Phi(\hat{p}, \epsilon; v) = \max_{p \geq 0, \mathbf{1}^\top p = 1, \|p - \hat{p}\|_1 \leq \epsilon} p^\top v. \quad (3)$$

Introducing positive and negative displacement variables gives a linear program. Its compact dual is

$$\min_{\nu, \lambda, \theta} \quad \nu + \hat{p}^\top \theta + \epsilon \lambda, \quad (4)$$

$$\text{s.t.} \quad \nu + \theta_c \geq v_c \quad \forall c, \quad (5)$$

$$-\lambda \leq \theta_c \leq \lambda \quad \forall c, \quad \lambda \geq 0. \quad (6)$$

Strong duality holds because the primal confidence set is nonempty and compact. The implementation also evaluates Eq. (3) independently by moving at most $\epsilon/2$ probability mass from low-score to high-score coordinates, respecting simplex boundaries.

2 Adaptive Confidence Envelope

Let $\mathcal{C}_s = \{p \in \Delta_K : \|p - \hat{p}_s\|_1 \leq \epsilon_s\}$ and define

$$\bar{A}(\hat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\hat{p}_s, \epsilon_s; v_{M,a,s}). \quad (7)$$

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Theorem 1 (Exact adaptive universal-attacker envelope). *On the event $\mathcal{E} = \{p_s \in \mathcal{C}_s \forall s, y\}$, $A(P, M) \leq \bar{A}(\hat{P}, M)$ simultaneously for every stochastic release channel M , including one selected from the same table. For fixed M , $\bar{A}$ is the exact supremum of $A(P, M)$ over $\mathcal{C}_1 \times \dots \times \mathcal{C}_G$.*

Proof. Using Eq. (2), take the supremum over the product confidence set. The assignment family is finite, so the supremum and assignment maximum commute. For a fixed assignment, the objective is additive in sources and the confidence set is a Cartesian product, so each source maximization separates and equals the support function in Eq. (3). This proves exactness for every fixed M . On $\mathcal{E}$, the true law is one feasible point in every confidence set, giving the upper bound. The statement is pointwise over the complete channel space, so it remains true after any measurable same-table selection of M . $\square$

For positive stratum count $n_{s,y}$ and fixed allocation $\delta_{s,y}$, the confirmation uses

$$\epsilon_{s,y} = \min \left\{ 2, \sqrt{\frac{2 \log((2^K - 2)/\delta_{s,y})}{n_{s,y}}} \right\}. \quad (8)$$

A union bound gives $\Pr(\mathcal{E}) \geq 1 - \sum_{s,y} \delta_{s,y}$. A zero-count stratum receives radius two and is covered deterministically.

3 Channel Capacity and Common-Shift Transfer

Define

$$\text{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \quad (9)$$

Lemma 2 (Exact arbitrary-shift capacity). $\text{Cap}_G(M) = \sup_{r_1, \dots, r_G \in \Delta_K} A(R, M)$.

Proof. Expand $A(R, M)$ by Eq. (2). For fixed assignment a , each source law r_s can concentrate all mass on a token maximizing $v_{M,a,s}(c)$. The source suprema separate and the finite maxima commute, yielding Eq. (9). $\square$

For two sources, uniform-prior Bayes accuracy is $(1 + \text{TV}(r_1 M, r_2 M))/2$. The diameter of the convex hull of channel rows is attained by two rows, hence $2 \text{Cap}_2(M) -$

$1 = \max_{c,c'} \text{TV}(M(c, \cdot), M(c', \cdot))$, the Dobrushin coefficient.

For common pre-release channel T , define

$$\rho_T(M) = \max_c \text{TV}((TM)(c, \cdot), M(c, \cdot)). \quad (10)$$

Lemma 3 (Approximate invariance transfer). *For every source experiment P ,*

$$A(P, TM) \leq \min\{\text{Cap}_G(M), A(P, M) + \rho_T(M)\}. \quad (11)$$

Proof. For source s , convexity of total variation gives $\text{TV}(p_s TM, p_s M) \leq \sum_c p_s(c) \text{TV}((TM)(c, \cdot), M(c, \cdot)) \leq \rho_T(M)$. The success probability of any attacker changes by at most this distance. Averaging over sources and maximizing the attacker gives the additive bound. The capacity bound follows by treating $p_s T$ as an arbitrary fine-token law and applying the preceding lemma. $\square$

4 Structured External Shift

For label y , suppose

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \quad t_y \geq 1 - \eta_y, \quad (12)$$

with common $T_y \in \mathcal{T}_y$ and common retained mass t_y across sources. For a polytope with finite extremes $\mathcal{V}_y = \text{ext}(\mathcal{T}_y)$, define

$$\begin{aligned} \bar{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y, a \in [G]^L} \sum_s \Big[\\ (1 - \eta_y) \Phi(\hat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) \\ + \eta_y \max_c v_{M,a,s}(c) \Big]. \end{aligned} \quad (13)$$

Theorem 4 (Transform-exact post-selection certificate). *On $\mathcal{E}$, simultaneously for every selected M and every external law in Eq. (12), $A(Q, M) \leq \bar{A}_y^X(M)$. For fixed M , the right-hand side is the exact supremum over the product confidence sets, transform polytope, allowed retained masses, and residual laws.*

Proof. Expand balanced Bayes accuracy using Eq. (2). For fixed T, t , and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are $\Phi(\hat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s})$ and $\max_c v_{M,a,s}(c)$. The latter is no smaller because every coordinate of $T v$ is a convex combination of coordinates of v . The maximizing retained mass is therefore $t = 1 - \eta_y$. The resulting confidence support is convex in T , so its maximum over a polytope is attained at an extreme. All remaining maxima are finite and commute. This gives exactness for fixed M ; pointwise coverage over all channels preserves the bound after same-table selection. $\square$

For a transform family that cannot be enumerated, let $B_y(M) = \min\{\text{Cap}_G(M), \bar{A}_y(\hat{P}, M) + \rho_{T_y}(M)\}$.

Corollary 5 (Capacity-transfer fallback). *On $\mathcal{E}$,*

$$A(Q, M) \leq (1 - \eta_y) B_y(M) + \eta_y \text{Cap}_G(M). \quad (14)$$

Proof. The retained experiment is bounded by Eq. (11), the adaptive envelope bounds its unknown reference term, and residuals are bounded by capacity. Since $B_y(M) \leq \text{Cap}_G(M)$, the largest value over $t_y \geq 1 - \eta_y$ occurs at $t_y = 1 - \eta_y$. $\square$

The exact confidence envelope is no larger than this fallback because it is the supremum over the same set. At zero confidence radius, it becomes the known-population external risk

$$\begin{aligned} \frac{1}{G} \max_{T \in \text{ext}(\mathcal{T}_y), a \in [G]^L} \sum_s \Big[(1 - \eta_y) p_{s,y}^\top T v_{M,a,s} \\ + \eta_y \max_c v_{M,a,s}(c) \Big]. \end{aligned} \quad (15)$$

used by the separately implemented synthetic evaluator.

5 Utility Certificate

Fix decoder $g : [L] \rightarrow [J]$, let $\ell_y(z) = \mathbf{1}\{g(z) \neq y\}$, and write $u_y = M \ell_y$. Define

$$\bar{E}_{s,y}(M, g) = \Phi(\hat{p}_{s,y}, \epsilon_{s,y}; u_y), \quad (16)$$

$$\kappa_y(M, g) = \sup_{T \in \mathcal{T}_y} \max\{0, \max_c ((TM - M)\ell_y)(c)\}, \quad (17)$$

$$U_y(M, g) = \max_c u_y(c). \quad (18)$$

Define the transform-exact value

$$\begin{aligned} \bar{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[(1 - \eta_y) \\ \Phi(\hat{p}_{s,y}, \epsilon_{s,y}; T u_y) + \eta_y \max_c u_y(c) \Big]. \end{aligned} \quad (19)$$

Corollary 6 (Transform-exact adaptive utility certificate). *On $\mathcal{E}$, every selected (M, g) satisfies $\text{Err}_{s,y}(Q, M, g) \leq \bar{E}_{s,y}^X(M, g)$, and the right-hand side is exact over the stated confidence ball and shift class.*

Proof. Repeat the transform-exact proof for bounded loss ℓ_y , without averaging over sources or maximizing an attacker assignment. $\square$

The capacity-transfer fallback is

Corollary 7 (Adaptive shifted-utility fallback). *On $\mathcal{E}$, every selected (M, g) satisfies*

$$\text{Err}_{s,y}(Q, M, g) \leq (1 - \eta_y) D_{s,y}(M, g) + \eta_y U_y(M, g), \quad (20)$$

where $D_{s,y} = \min\{U_y, \bar{E}_{s,y} + \kappa_y\}$.

Proof. Apply the structured-shift proof to the bounded loss ℓ_y . The common component differs from its reference expectation by at most the largest positive directional defect κ_y , while the arbitrary residual concentrates on a row with maximal error U_y . $\square$

The transform-exact utility value is never larger than this fallback.

The exact population utility risk used in the independent evaluator is

$$\max_{T \in \text{ext}(\mathcal{T}_y)} \Big[(1 - \eta_y) p_{s,y}^\top T M \ell_y + \eta_y \max_c (M \ell_y)(c) \Big]. \quad (21)$$

6 No-Free-Lunch Boundary

Theorem 8 (Unrestricted differential shift). *For $G \geq 2$, $\text{Cap}_G(M) = 1/G$ if and only if all rows of M are identical.*

Proof. Identical rows make every released source law identical. Conversely, if two rows differ, choose two residual source laws concentrated on those rows and place every remaining source on either row. The released source laws are not all identical, so uniform-prior Bayes accuracy is strictly above chance. Exactness of Eq. (9) completes the argument. $\square$

For two fine tokens c_0, c_1 with different task labels, suppose a decoder has row-wise errors at most e_0, e_1 . The decoder decision event occurs with probability at least $1 - e_0$ under $M(c_0, \cdot)$ and at most e_1 under $M(c_1, \cdot)$. Total variation is therefore at least $1 - e_0 - e_1$, giving

$$2 \text{Cap}_2(M) - 1 \geq 1 - e_0 - e_1. \quad (22)$$

Thus a nontrivial task contract is incompatible with chance-level source capacity under unrestricted differential shift.

7 Missing-Source Non-Identifiability

Fix two required sources and a released-token law $q_0 M$ for source zero. Suppose an external audit contains no source-one rows, so its observed-data distribution is identical for every $q_1 \in \Delta_K$. Define

$$A_{\text{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M). \quad (23)$$

Theorem 9 (Missing-source non-identifiability). *Let a possibly randomized protocol observe no source-one rows and claim $A((q_0, q_1), M) \leq \tau$. If its false-certification probability is at most $\delta < 1$ uniformly over every q_1 , then for $\tau < A_{\text{miss}}(q_0, M)$ it can issue that claim with probability at most δ .*

Proof. The finite-simplex supremum is attained; call a maximizer q_1^* . The audit's observed-data distribution does not depend on the missing law. Consequently, every event on which the protocol claims a bound below $A((q_0, q_1^*), M)$ is a false-certification event under q_1^* , and uniform validity limits that event to probability δ . $\square$

The same indistinguishability argument for a missing source-label utility stratum replaces Eq. (23) by $\max_c (M \ell_y)(c)$. These are precisely the full-simplex terms used when a stratum has count zero. Moreover, if M is non-constant, then $A_{\text{miss}}(q_0, M) > 1/2$ for every q_0 : the mixture $q_0 M$ cannot equal every nonidentical channel row, and a missing source concentrated on a differing row has positive total-variation distinguishability. Missing required strata must therefore be reported as unestimable unless an external structural assumption identifies their laws.

8 Abstention Theory

Let $V(P, e)$ be the globally optimal certified worst conditional error for confidence balls centered at P with common radius e . With deployed radius e_n , define $r_n^* = \sup\{r \geq 0 : V(P, e_n + r) \leq \tau_U\}$.

Proposition 10 (Finite-sample abstention envelope). *For H registered strata of size n ,*

$$\Pr(\text{MOSAIC abstains}) \leq \min\{1, H(2^K - 2)e^{-n(r_n^*)^2/2}\}. \quad (24)$$

Proof. If every empirical stratum is within distance r of its population law, each radius- e_n ball around the empirical law is contained in the radius- $(e_n + r)$ ball around the population law. A feasible population-centered pair for the larger balls remains feasible for the empirical problem. Apply the Weismann bound and a stratum union bound, then let r approach r_n^* . $\square$

For the preregistered local power curve, write $V_n(P) = V(P, e_n)$. Under local uniqueness of the decoder and certificate branches together with strong regularity of the resulting parametric linear program, the value is differentiable. If $h_{n,s,y}$ is its simplex gradient and $\Sigma_{s,y} = \text{diag}(p_{s,y}) - p_{s,y} p_{s,y}^\top$, the finite- n delta-method prediction is

$$\Phi_N \left(\frac{\tau_U - V_n(P)}{\sqrt{n^{-1} \sum_{s,y} h_{n,s,y}^\top \Sigma_{s,y} h_{n,s,y}}} \right). \quad (25)$$

The locked analysis estimates each gradient by centered simplex-preserving finite differences at step sizes 10^{-4} and 5×10^{-5} and rejects the approximation if their maximum discrepancy exceeds 5×10^{-3} .

9 Optimization Certificate

For fixed decoder, each support-function dual, transformed score, and residual row maximum in Eqs. (13) and (19) has a linear epigraph in the channel and auxiliary variables. Enforcing all transform extremes and attacker assignments therefore gives one linear program per decoder; enumerating all J^L decoders returns the global transform-exact finite-alphabet optimum. The capacity-transfer fallback represents each minimum as a binary disjunction and solves a mixed-integer linear program per decoder.

Globality is asserted only when every decoder and attacker assignment is enumerated, the exact LP reports an optimal status or the fallback reports numerical zero MIP gap, the primal residual is at most 2×10^{-7} , and an independent implementation reproduces every certificate and the objective at the same tolerance. A timeout, nonzero gap, oversized assignment family, or post-hoc mismatch cannot return a deployment.

The exact audit checks zero-radius agreement with a separate population evaluator, binary confidence-ball endpoints, 100 randomized dominance comparisons with the fallback, and dense channel grids. The fallback audit compares 30 randomized small instances with exhaustive 21^K grids and includes a witness in which the best stochastic certified error is 0.371588 while every deterministic release has certified error 1.

10 Locked Experimental Protocol

The synthetic population has two labels, two sources, and three fine tokens. For label zero the source laws are (0.80, 0.15, 0.05) and (0.65, 0.30, 0.05); for label one they are (0.05, 0.20, 0.75) and (0.05, 0.35, 0.60). The common-transform set is the convex hull of identity and a nearest-neighbor smoothing channel. The hard safety scenario uses $\eta = 0.20$, privacy-advantage threshold 0.30, and worst conditional error threshold 0.40. The retention scenario uses $\eta = 0.10$, thresholds 0.35 and 0.45. Every source-label stratum contains exactly n independent draws.

The confirmation fixes 1,000 replicates in each of three hard-scenario and five retention-scenario sample-size cells. It compares eight rules and stores 64,000 method-level receipts. The preregistration fixes all seeds, methods, intervals, and pass criteria, hashes the theorem and implementation, and was pushed before the confirmation began. Pilot seeds zero through 199 are excluded. The oldest aggregate-only pilot is retained as design history but cannot support a channel-level claim; every later pilot row is replayed by the exact evaluator.

11 Confirmation Receipts and Complete Primary Cells

The locked preregistration has SHA-256 digest
d6819cec1605db6dfef5789c0f0f8d67
4ee135b51f1cd820aaff16df84b65cf2. The 45 MB confirmation receipt has digest
de4fd53da17e0371d9023ef5a58f2cf3
093ed3111b980920e0f9b9eab1b4e665. Its separate replay checks all 64,000 seeds and channels, 192,000 stochastic rows, 128,000 exact risks, 128,000 decision labels, and 1,152 aggregate fields; the maximum replay discrepancy is exactly zero in double precision. The replay itself has digest
9e64b4883cfb5bf510a058e0fe6995abb
e362b2d958c0d1163fbaca78d2966d4.

For zero events in 1,000 trials, the exact 95% Clopper-Pearson upper endpoint is 0.3682%. MOSAIC's retention rate has interval [53.16, 59.40]; held-out certification has [1.62, 3.67] and finite LTT has [2.28, 4.60]. The theory alignment audit links both pre-outcome and post-outcome receipts and passes with mean absolute deployment error 0.001968 and primary-cell error 0.004389. The finite-sample concentration abstention bound is vacuous through $n = 500$, then falls to 0.03097 at $n = 1000$ and 7.60×10^{-9} at $n = 2000$; we report that looseness rather than replacing it with the local approximation.

12 Transform-Exact Confirmation and Audit Amendment

The strict-decision v2 preregistration has SHA-256 digest
89243fea30a158213679ed310e232e97c
874137a57d97cccd4f6f07c6ce4b867. Its full 10,000-row report has digest
80c5c77a5d98eba16a4a9a5d60f38871
8115ef600546bc766aec106eebf1421, and the independent audit digest is

b75a99e90b8e28e042cf2b1e5ec0ce72
651dc44f0b015b0b7c6cacda71ce78be.

Declared numerical amendment. The original v1 confirmation passed every pre-audit gate, but its independent replay marked two of 10,000 deployment decisions as mismatches. All 80,000 recomputed certificate and external-risk quantities matched. The two returned capacity-transfer channels exceeded the privacy threshold by 4.12×10^{-10} and 4.40×10^{-10} ; below the solver's 10^{-9} feasibility and 2×10^{-7} post-hoc tolerances, but above the replay's 10^{-10} decision tolerance. We preserved the failed audit, changed the deployment layer to recheck every privacy certificate at 10^{-10} , and recorded before rerunning that exactly those two rows should flip to abstention without changing any primary gate. The hash-locked v2 rerun matched that prediction exactly; its audit has zero mismatch.

The confidence event holds on 999 of 1,000 tables in the hard cell, on all retention $n = 125$ tables, on 999 tables at $n = 250$ and 500, and on all tables at $n = 1000$. There are no failures on the confidence event and no observed false acceptances off it. For zero events in 1,000 trials, the exact 95% binomial upper endpoint is 0.3682% per method-cell; this empirical endpoint is distinct from the theorem's familywise $\delta = 0.05$ guarantee.

The exploratory five-threshold receipt has digest
5147a0a73ce888265351baa8ee638b75
1fc4837b60f96335d1fccd14442bdeea. It reuses the audited channels and exact population risks, holds the privacy threshold at .35, and changes only the utility decision threshold. Because four thresholds were selected after outcomes, the sensitivity table in the main paper is explicitly descriptive. All 20 method-size-threshold cells have zero false acceptance, and exact retention is never lower than capacity transfer.

13 Transform-Exact Verification Matrix

The exact support implementation is checked in three independent ways. At zero radius, privacy and utility match a population evaluator that enumerates every transform extreme, attacker assignment, and residual simplex vertex. For binary fine alphabets, confidence-ball endpoint enumeration reproduces the support-function result. Finally, 100 randomized instances verify that the exact envelope never exceeds the capacity-transfer fallback. The optimizer is then checked against dense channel grids and an independent post-solve certificate recomputation. Eleven focused transform-exact tests pass as part of a repository-wide run of 154 tests plus 14 subtests.

14 Official-Method Real-Feature Receipts

The real-feature preregistration digest is
362065c8f110fd3eb096a638e12031ef
494eb208524c69216910307033dd7ed3. The complete manifest digest is
773ba872cab2b404900652a3d8f8f522
e9e29295df2d50bd2736d109e7182b0b; its independent replay digest is
bbf30f5db4f466325063ef0edb8c282b

Rule	Hard safety boundary, $n = 125$			Retention regime, $n = 250$		
	Deploy	False	Safe	Deploy	False	Safe
Always deploy plug-in	100.0	57.6	42.4	100.0	0.0	100.0
Plug-in continuum	73.7	42.1	31.6	100.0	0.0	100.0
Shift-unaware MOSAIC	80.3	76.6	3.7	100.0	0.0	100.0
Held-out fixed	0.0	0.0	0.0	2.5	0.0	2.5
Finite LTT	0.0	0.0	0.0	3.3	0.0	3.3
Deterministic MOSAIC	0.0	0.0	0.0	0.0	0.0	0.0
MOSAIC	0.0	0.0	0.0	56.3	0.0	56.3
Population oracle	100.0	0.0	100.0	100.0	0.0	100.0

Table 1: All eight locked rules in both primary cells (rates in percent). The oracle verifies that a safe channel exists; it is not deployable.

Scenario	n	Safe deploy (%)		Mean cert. error		Mean gain (pp)	False accepts
		transfer	exact	transfer	exact		
Hard boundary	125	0.0	0.0	.5004	.4959	0.45	0/2000
Retention	125	0.3	53.0	.4808	.4482	3.26	0/2000
Retention	250	57.8	99.9	.4479	.4070	4.09	0/2000
Retention	500	100	100	.4162	.3799	3.63	0/2000
Retention	1000	99.8	100	.3888	.3612	2.77	0/2000

Table 2: Every locked transform-exact confirmation cell. Each method has 1,000 paired tables per row. The exact objective is no larger on all 5,000 pairs and strictly smaller on all 4,000 retention pairs.

Seed	Selected	Cert.	Ext. error	Ext. adv.
100	INLP rank 4	.3124	.1817	.0451
101	R-LACE rank 4	.3086	.2030	.0720
102	INLP rank 4	.3259	.2047	.0843
103	INLP rank 1	.2991	.1954	.0598
104	INLP rank 4	.3160	.1863	.0456

Table 3: BiasBios selected rows; all are externally safe.

Seed	Selected	Cert. error
100	LEACE	.2702
101	LEACE	.2818
102	LEACE	.2773
103	LEACE	.2955
104	LEACE	.2853

Table 4: Camelyon17 selected rows. Every natural external diagnostic is unestimable because a required source is absent.

47916b287c9139b6595b8b14b86af4d6. All seven locked gates pass: complete execution, positive usefulness, empirical safety, support discipline, modality coverage, claim discipline, and receipt replay.

The frontier contains 13 rows in a fixed order: identity; INLP ranks 1, 2, 4, and 8; LEACE; R-LACE ranks 1 and 4; TaCo with 1, 2, 3, and 5 removed components; and MANCE++. Every transformation calls a pinned official implementation and records its repository, commit, entry point, and parameters. There are no proxy rows. Each candidate has a separately fitted tokenizer, so its simultaneous table receives $\delta/13$ before continuum channel selection. Because the protocol predates the transform-exact amendment, these rows use the capacity-transfer fallback. The official-method study therefore evaluates the frontier and release protocol, while the paired synthetic study measures the exact refinement itself.

All 65 BiasBios candidate rows are certifiable and externally safe. By contrast, all 195 rows across Waterbirds, CivilComments, and GaitPDB satisfy the privacy threshold only with certified worst error above .49; the selected decision is therefore abstention. Natural external evaluation finds zero safe candidates in CivilComments and GaitPDB and one of 65 in Waterbirds. That Waterbirds row is a useful conservative case: favorable external behavior does not retroactively

make an uncertified construction-table decision deployable.

Camelyon17 has 65 certifiable candidate rows and selects LEACE on every seed, but every external row is missing a required source class. The external audit distribution is therefore identical under arbitrarily different missing-source laws, exactly matching the non-identifiability theorem. We count these five model-conditional releases separately from the five externally estimable BiasBios selections. Reporting “zero false acceptances” refers only to those five estimable selections and is not promoted to a population-frequency claim.

Exploratory transform-exact replay. After confirmatory outcomes were known, we re-optimized the exact envelope on all 325 stored token tables. Exact certified error is no larger on every row and strictly smaller on 133, with a maximum reduction of 3.54 percentage points. Candidate deployments increase from 130 to 132: INLP rank 4 on Waterbirds seeds 102 and 103 crosses the utility contract, and both selected releases pass the natural external diagnostic. Exact selection therefore deploys on 12 of 25 jobs; all seven externally estimable selections are diagnostically safe, five Camelyon17 selections remain unestimable, and the remaining 13 jobs abstain. This analysis is post-outcome and exploratory, not a new confirmatory study.

The analysis digest is
ea45e4ed7c47691e1813fea40d36293e
295db03245c11f4485c339ae7778aeae; its
independent 325-optimization replay has digest
3f3afabf41d5e4ff0c28c5d21dd68813
af54c6b47ea2f0e9d478bd9bd92850ac and zero
mismatches.

15 Reproducibility Manifest

MOSAIC optimization, confirmations, and audits ran on an Apple M4 system with 10 CPU cores and 16 GB unified memory under macOS 26.2 (arm64). The locked Python environment was Python 3.12.13 with NumPy 2.5.1, SciPy 1.18.0, scikit-learn 1.9.0, PyTorch 2.13.0, and the HiGHS LP/MILP backend exposed by SciPy. No GPU is required once the registered feature stores have been constructed; synthetic confirmation and every replay are CPU-only.

The named package reports the exact preregistration hash, Git commit, software versions, and artifact hashes. The anonymous package retains the same internal identifiers while replacing public authorship and repository metadata as required by double-blind review. Raw synthetic receipts contain no private or human-subject data. Real-domain experiments, if included, retain dataset terms, official split definitions, preprocessing hashes, and method-specific code provenance in a separate manifest.